

Outlines

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Out line of the course (Around 13-15 hrs)

- I presume that you have fair knowledge on basic mathematics
- Signals and systems introduction/review
- Continuous time signals
- Discrete time (DT) signals and systems in time domain
- Typical processing of DT signals with DT systems
- LTI systems
- Characterization and classification of LTI systems
- Sampling in time, aliasing, interpolation, and quantization.
- Basic operations like convolution
- DT signals in frequency domain
- DFT and DTFT and convolution using these transforms
- Elementary Z-transform
- LTI DTS in frequency domain
- Transfer function, frequency response
- Simple digital filters, characterization and design
- Sampling and reconstruction
- Realization of digital filters
- FIR and IIR filters, basic properties of DT systems.
- Correlation in time and frequency domain

Intro. To Signal and Wave

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- **WAVE** is a movement in time in a medium. **WAVE** needs medium
- Signal contains information, and its variations and represented in the form of waves
- SIGNAL information is about the state or behavior of a physical system

Signal is just of variation of the behavior of the state of a physical system.

Signal → $Y = f(x)$, x = time, temp. density, pressure, etc...

Example:

- Variations of acoustic pressure in speech signal
- Variations of reflectance value in a picture/image
- Variations with depth of physical quantities such as density, porosity, and electrical resistivity in Geophysics
- Variations of air pressure, temperature, and wind speed with altitude in meteorological investigations
- Variations of average budget, crime rate, or pounds of fish caught in demographic studies,
- etc.

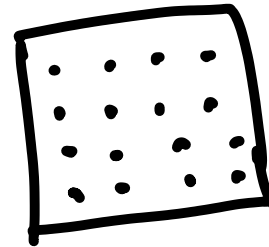
Types of Signal

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Type:

- Natural & Synthetic
- In terms of dimensions
- In terms of sources
- Continuous and Discrete
- Analog and Digital

$$x = f(t) \rightarrow 1D$$
$$x = f(t_1, t_2)$$



- SIGNAL is a function of time, but in DSP, it lost its significance
- All CT signals are analog, but not the other way.

Why Signal and its Processing?

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• Why Signal?

- A signal contains information

• Why Processing?

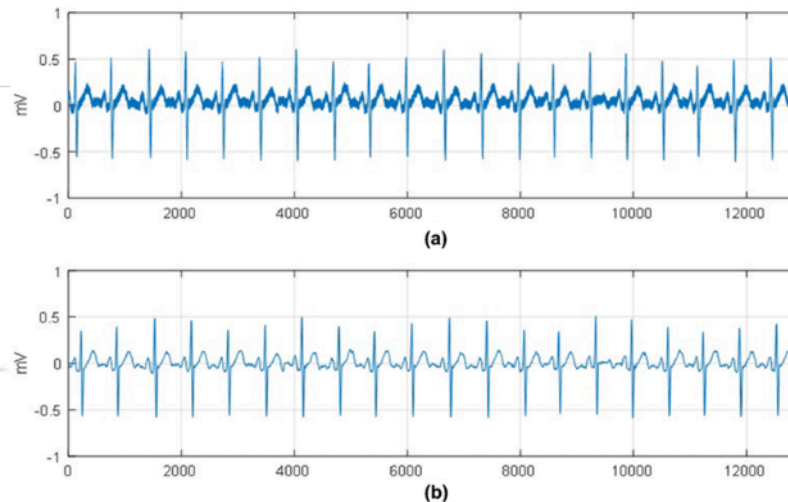
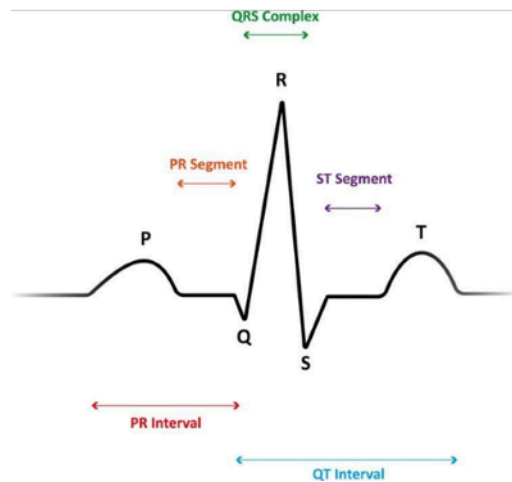
- Signal has information is not sufficient

• Example of processing

- Modulation, demodulation
- Multiplexing, demultiplexing
- Filtering, and so on.

Corruption:

- 50hz power-line pick up (induction)
- Electromyography (EMG) signals



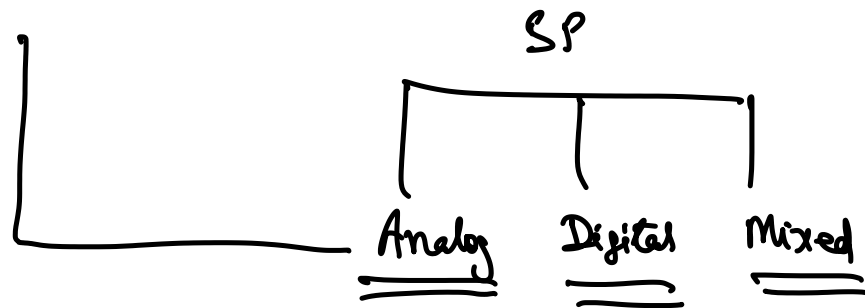
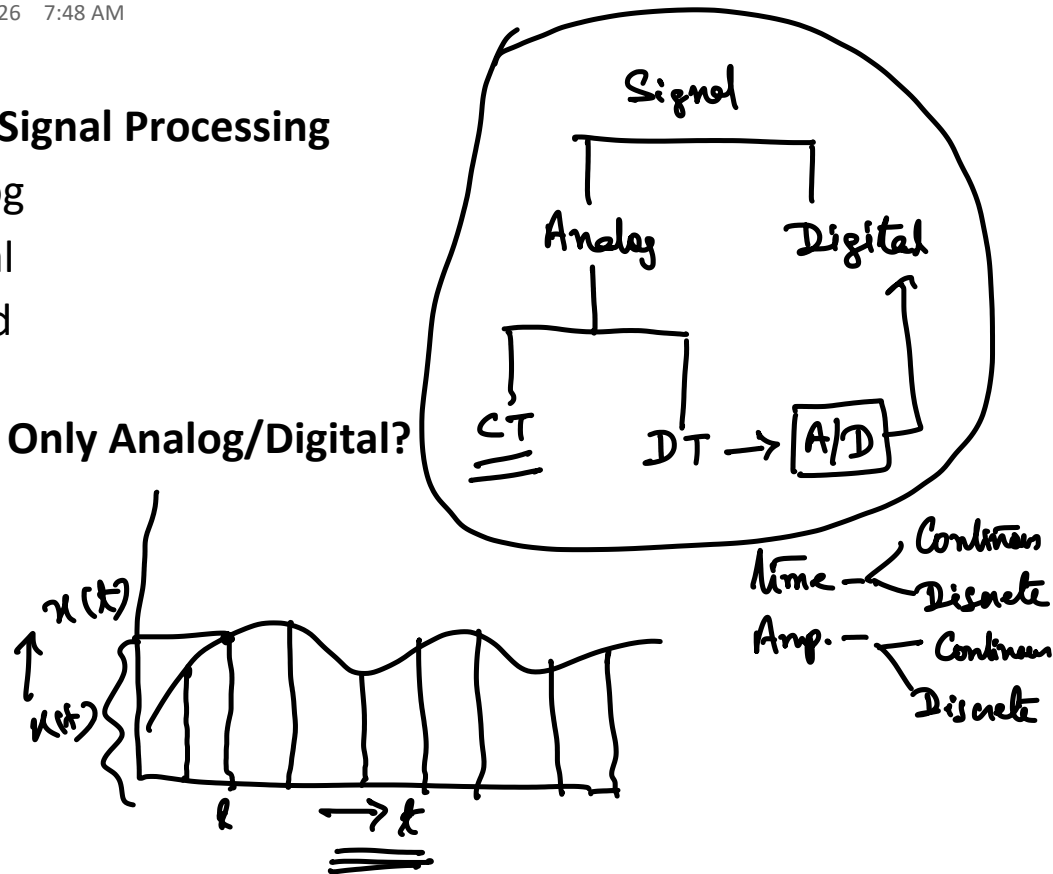
Ways to perform SP

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Ways of Signal Processing

- Analog
- Digital
- Mixed

Why not Only Analog/Digital?



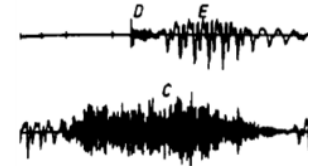
Math Rep. of Signal

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SIGNAL:

A signal is a function (f) that maps an independent variable to a value (**amplitude**):

$$\underline{\underline{x = f(t)}}$$

Speech Signal:**Examples of Signals mathematically:**

1) Linear Signal $x(t) = 2t + 4$

2) Quadratic Signal $x(t) = t^2 + 1$

Exponential = e^t

$$\boxed{\text{Sinusoidal Signal} = A \sin(\omega t + \phi)}$$

$$\sum_{i=1}^N A_i(t) \sin[2\pi F_i(t)t + \theta_i(t)]$$

$$S_1(t) = 2t$$

$$S_2(t) = 12t^2$$

$$\boxed{S_1(t) + S_2(t) = 2t + 12t^2}$$

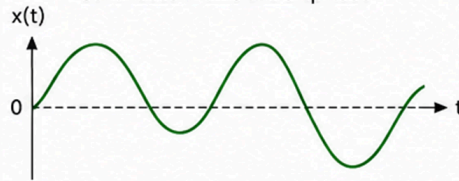
$$S(x, y) = 3x + 10xy + 2x^2y$$

Signal Categorization

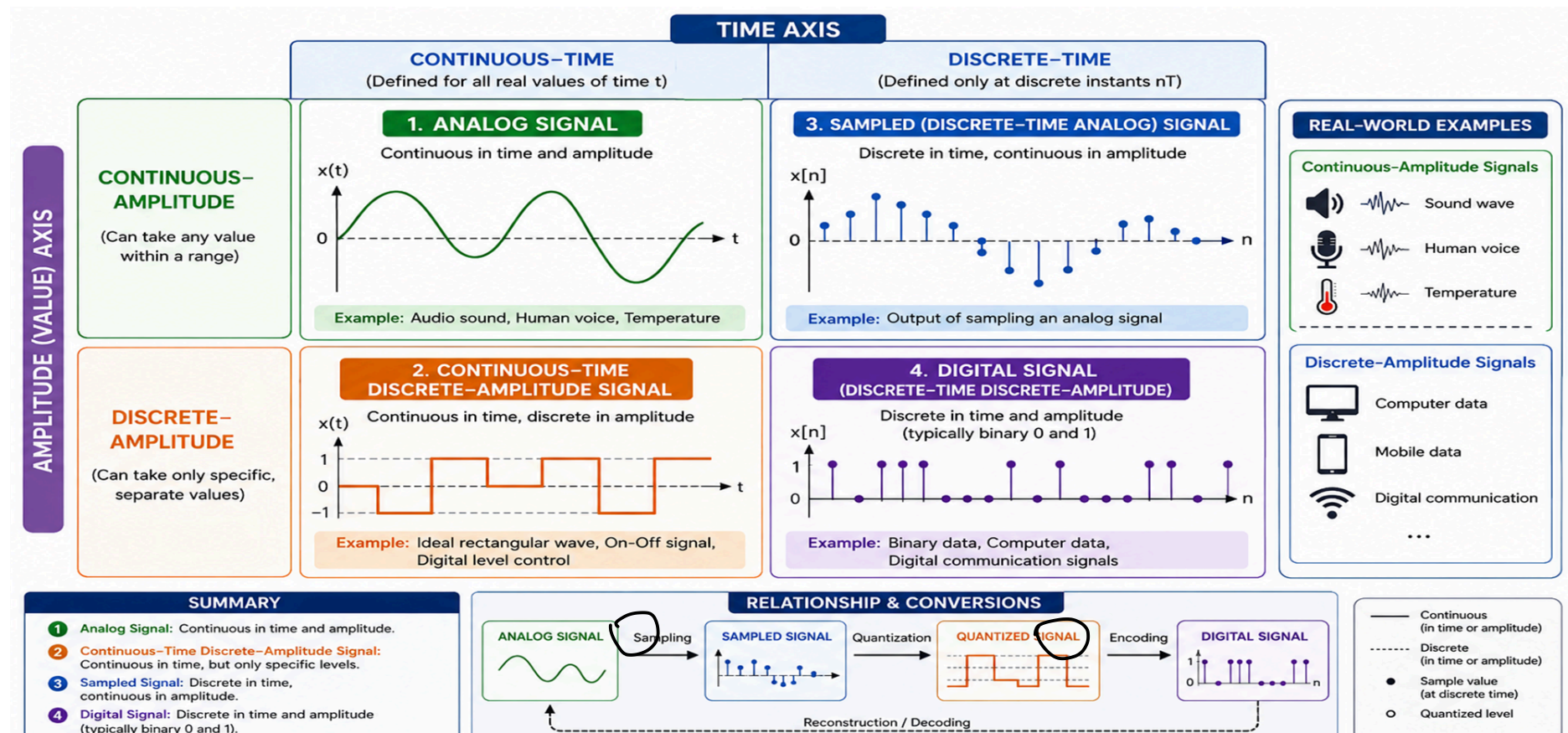
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1. ANALOG SIGNAL

Continuous in time and amplitude

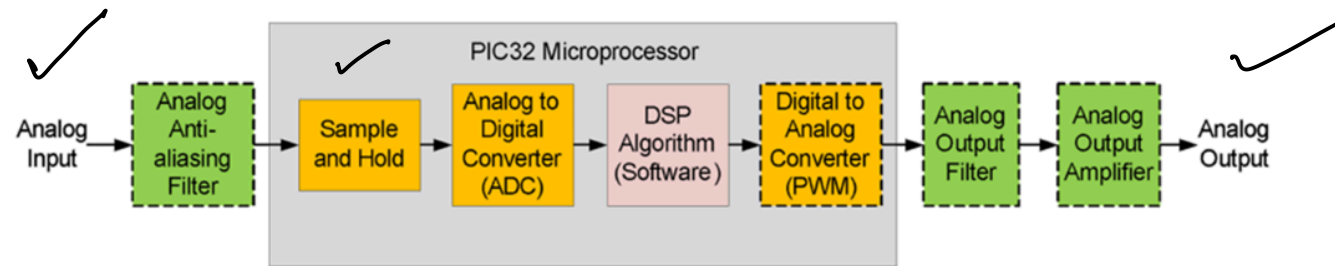
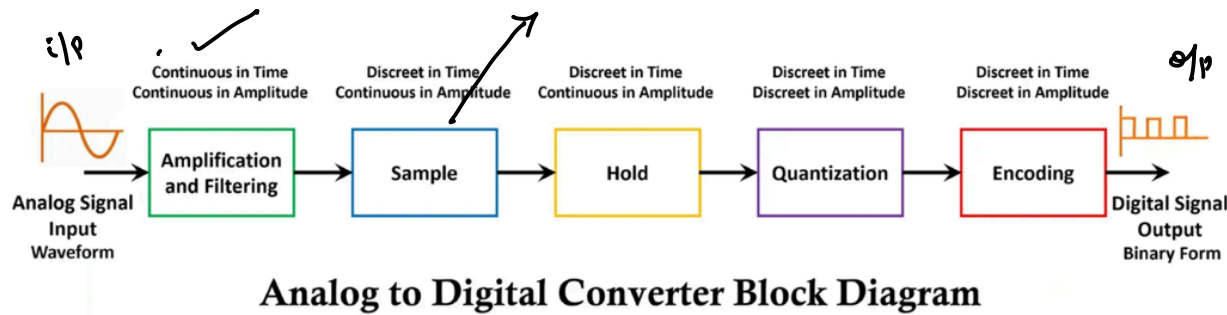


Example: Audio sound, Human voice, Temperature



Analog => Digital => Analog

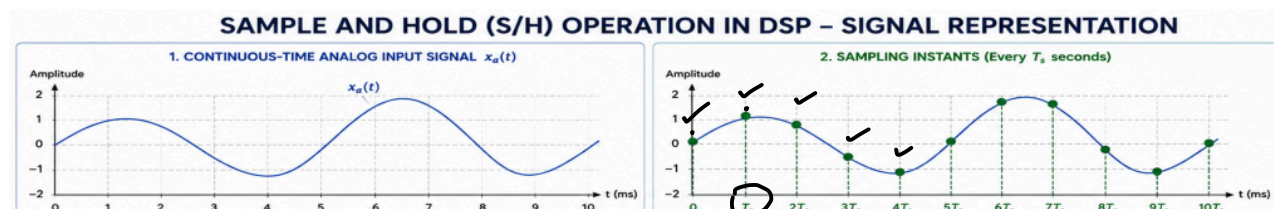
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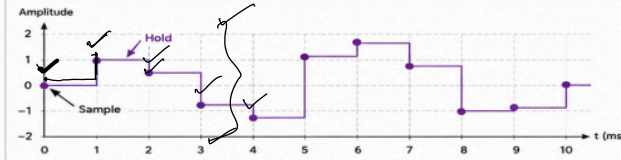


Digital Signal Processing

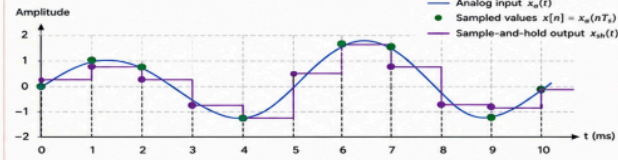
$$x_a(t) \rightarrow \frac{x_a(nT)}{T_s} \Rightarrow x(n) \Leftarrow$$

$$x_s(t) \rightarrow \boxed{\text{Sample}} \downarrow$$

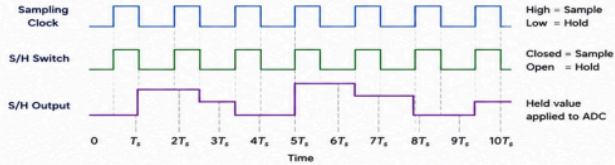


3. SAMPLE AND HOLD OUTPUT $x_{sh}(t)$ (STAIRCASE WAVEFORM)

4. RELATION BETWEEN SIGNALS



5. TIMING DIAGRAM

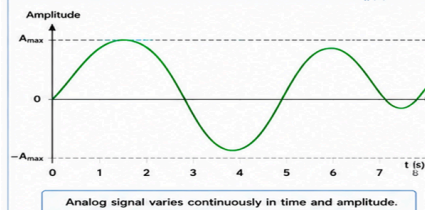


10 ms, 15 mV, 20 mV

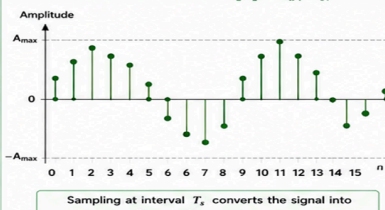
$f_s \neq 2f_0$

QUANTIZATION AND ENCODING OF SIGNALS IN DSP

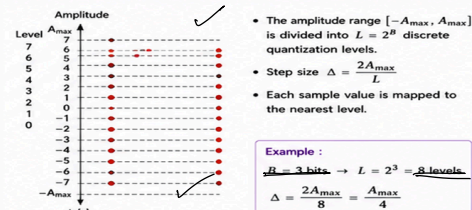
From Continuous-Time Signal to Digital (Encoded) Signal

1. CONTINUOUS-TIME ANALOG SIGNAL $x_a(t)$ 

Analog signal varies continuously in time and amplitude.

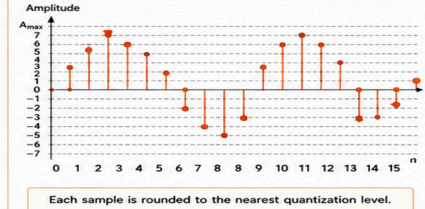
2. SAMPLED SIGNAL $x[n] = x_a(nT_s)$ Sampling at interval T_s converts the signal into discrete-time sequence.

3. UNIFORM QUANTIZATION



- The amplitude range $[-A_{max}, A_{max}]$ is divided into $L = 2^B$ discrete quantization levels.
- Step size $\Delta = \frac{2A_{max}}{L}$
- Each sample value is mapped to the nearest level.

Example :
 $B = 3 \text{ bits} \rightarrow L = 2^3 = 8 \text{ levels}$
 $\Delta = \frac{2A_{max}}{8} = \frac{A_{max}}{4}$

4. QUANTIZED SIGNAL $x_q[n]$ 

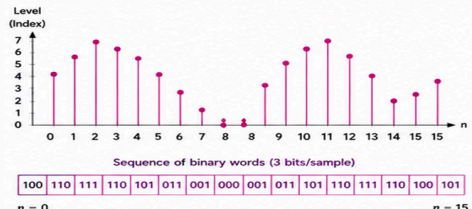
Each sample is rounded to the nearest quantization level.

5. ENCODING (BINARY REPRESENTATION)

Level Index	Quantization Level	3-bit Binary Code (B = 3)
7	$+7\Delta/2$	111
6	$+5\Delta/2$	110
5	$+3\Delta/2$	101
4	$+1\Delta/2$	100
3	0	011
2	$-1\Delta/2$	010
1	$-3\Delta/2$	001
0	$-5\Delta/2$	000

- Each quantized level is represented by a B-bit binary word.
- Number of levels: $L = 2^B$
- Number of bits per sample: B

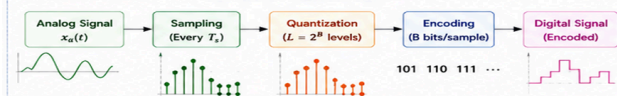
Example :
 $B = 3 \text{ bits} \rightarrow L = 8 \text{ levels}$
 Codes range from 000 to 111

6. ENCODED DIGITAL SIGNAL $b[n]$ 

Sequence of binary words (3 bits/sample)

100 110 111 110 101 011 001 000 001 011 101 110 111 110 100 101
 $n = 0$ $n = 15$

COMPLETE PROCESS FLOW



KEY POINTS

- Quantization converts a continuous amplitude range into a finite number of discrete levels.
- The resolution of quantization depends on the number of bits (B).
- Higher B $\rightarrow$ more levels $\rightarrow$ lower quantization error $\rightarrow$ better accuracy.
- Encoding converts each quantized level into a binary code for digital processing, storage, and transmission.

